

Influence of Community Health on U.S. Migration

Freya Van de Motter, Ransom MacLeith, and Shay Gooding

Background

Relative to previous generations, modern migration decisions are less influenced by traditional life events, such as job relocation, home ownership, and starting a family. This enables greater influence from a diverse set of possible factors when deciding to migrate.

Addressing all possible factors is outside the scope of this project; we decided to focus on community health. Existing research on British households between 1991 and 2008 concludes (1) that migration impacts personal health and (2) that health possibly motivates migration aspirations¹.

To help us explore something similar, we will use the University of Wisconsin's County Health Rankings aggregates more than 75 metrics on population health and community conditions in an effort to better understand the differences in health across US communities².

Sources: (1) [Internal migration, area effects, and health](#), (2) [County Health Rankings and Roadmaps](#)

Motivation

Most existing health-migration research in the US focuses on post-migration health effects³. We would like to do the opposite, exploring the extent to which health conditions predict migration decisions. In addition, the study on British households used mortality as a proxy for health. In contrast, the County Health Ranking's wealth of aggregated metrics enables a more nuanced and specific analysis of what health factors are most influential.

Furthermore, we all have vested personal interest in migration across the U.S. Freya has lived in nine counties across two continents and four US States. Shay has recently migrated to Texas from California. And Ransom will be moving when he leaves the Army this summer. To some extent, our past moves have been made for us; we're interested in learning how to better inform ourselves in preparation for migration decisions that will be more of our own.

Source: [Migration Decision-Making and its Key Dimensions](#)

Objectives

We aim to identify basic and advanced insights about migration, population health, and community conditions across the US at the county level.

We will examine migration rates by county, region, and state, and look for geographic patterns in overall county health.

Then we'll dig into those datasets together. We'll see if people are moving into healthier communities and out of unhealthier ones, or if other factors may be leading to otherwise counterintuitive migration patterns. We'll explore the data visually and statistically.

Finally, we'll perform regression analyses to see how much of the variation in migration (in, out, and net) can be attributed to the metrics in our datasets, and what we can learn from the results.

Data Sources

We leveraged three datasets from two institutions for our analysis

ACS Migration Flow

Our primary dataset is the US Census Bureau's American Community Survey (ACS) 5-Year Migration Flow 2016-2020. It includes data about migration, with migration primarily measured by residents' survey responses to where they lived when surveyed and where they lived one year ago. The data is collected over a 5-year period to ensure larger samples for smaller-population counties and then used to produce an annual average.

Estimated size: Approximately 480,000 records (~406k county-to-county records).

Location: We called an API using Python 'Requests' to retrieve the data in JSON format. Information about the API can be found here: <https://www.census.gov/data/developers/data-sets/acs-migration-flows.html>

County Health Rankings

Our secondary dataset is the 2025 County Health Release National Data from the University of Wisconsin Population Health Institute (UWPHI). This dataset lists z-scores for Population Health & Well-Being (PHWB) and Community Conditions (CC) by county, based on ~30 underlying measures. These summaries were only developed in 2024 and 2025, so we chose 2025.

Estimated size: ~3210 records (~one row per US County) with many columns (each attribute and related descriptive statistics (confidence interval, etc), Population Health and Wellbeing z-score, Community Conditions z-score, etc.) with additional (and optional) comprehensive columns for numerous health factors.

Location: [2025 County Health Release National Data](#) - Excel file we downloaded.

County Health Metrics

We added a tertiary dataset from UWPHI's County Health Rankings, the 2022 Analytic Data. We chose 2022 because the underlying data was largely gathered 2016-2020, overlapping our migration data¹.

We used 74 'raw value' feature columns. These included health metrics like excessive drinking and physical inactivity, demographic data like race, age, gender, and economic data like food insecurity and broadband access.

Estimated size: 3196 rows (3194 counties) and 725 columns in a CSV loaded from their site using `pd.read_csv()` from our notebook.

1: [See the County Health Rankings 2022: Analytic Dataset Codebook](#)

Location: [County Health Rankings Data & Documentation \(2022\)](#)

Data Manipulation Methods

Overview of the data manipulation methods used on our primary datasets

County Health Rankings

Initial Shape: (3210, 10)

- Dropped 51 state header rows
- Dropped 70 rows with missing population z-scores.
- Converted z-scores to a numeric data type.
- Multiplied z-scores by -1 so that higher scores equated to better health outcomes.
- Renamed columns of interest to Health_Z_Score and Community_Z_Score.
- Keep only: FIPS, State, County, Health_Z_Score, and Community_Z_Score
- Used `zfill(5)` and forced data type object to match 5-digit FIPS code with those of other datasets in preparation for dataframe merge.

Final Shape: (3089, 5)

County Health Metrics

Initial Shape: (3196, 725)

- Used regex to select 74 raw-value feature columns by their `'_rawvalue'` suffix
- Use a list to include 5 “identifier” columns
- Pivoted the two header levels to create a “config” file that mapped the second-level headers to the more human-readable top-level.
- Identified missing values. 30/74 columns had missing values. 8/74 had missingness of more than 5% but less than 20%.
- Imputed missing values with Scikit-Learn’s `IterativeImputer`.
- Applied regex on feature names when plotting values to make visualizations more readable.

Final Shape: (3142, 79)

ACS Migration Flows

Initial Shape: (480130, 13)

- Applied `zfill(5)` to convert FIPS code to dtype object as merge common variable.
- Asserted no missing values (US Census Bureau interprets no reported migration as zero county-to-county migration and excludes these county pairs from the dataset).
- Removed state- or country-level migration flows by filtering to only five-digit FIPS codes
- Mapped generic headers like `FULL1_NAME` and `GEOID1` to semantically meaningful names like `Origin_County_Name` and `Origin_FIPS_Code`.

Final Shape: (406348, 13)

Health-Migration Merges

- Applied `groupby().agg()` to ACS Migration flows to create a dataframe with one row per county, taking the sum of in-, out-, and net-migration as well as the mean of county population
- Calculated migration rates (people per 1000) for each county
- Used inner join to merge rankings and migration on FIPS code.

Final Shape: (3080, 21)

State-Level Analysis

- Applied `groupby().agg()` to county migration data to create state-level summaries of health and community z-scores. Z-scores were weighted by county population when aggregating.
- Calculated state migration rates.

Final Shape: (50, 6)

Final Merge: County Health Metrics + Health-Migration by County inner joined on FIPS code. **Final Shape: (3080, 91)**

Data Manipulation Methods and Analysis

County Health Ranking Dataset Distributions

Distribution Plot: We used tidy data transformation using `pandas.melt` and pivoted the dataframe from two columns to a single column for the values (Score) and a categorical column (Metric). We chose `seaborn.displot` to normalize the Health and Community Conditions distributions in a shared density visual.

1. Health Z-Score (Orange): This is weighted heavily by mortality:

- **50%:** Premature Death.
- **50%:** Low Birthweight (20%), Poor Physical Health Days (10%), Poor Mental Health Days (10%), and Poor/Fair Health (10%).

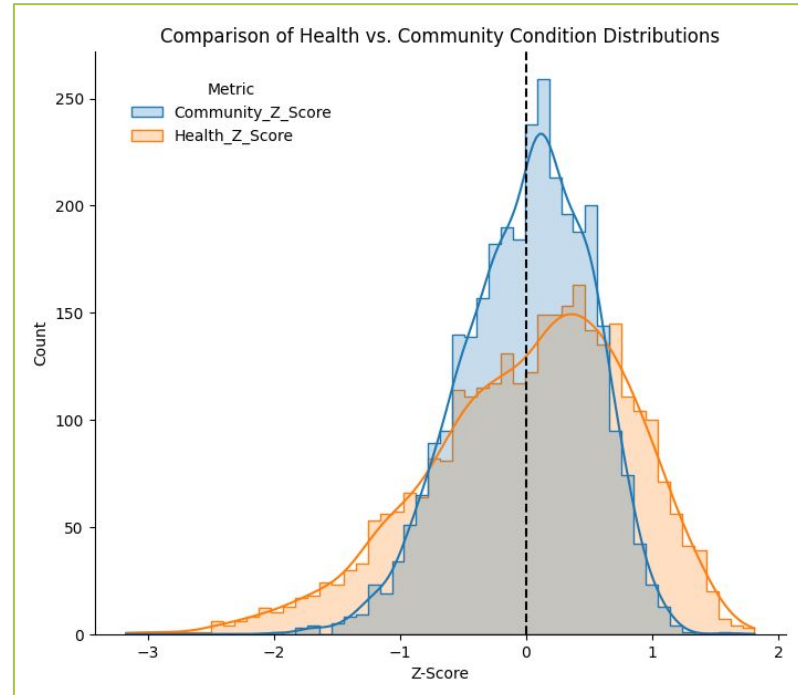
Variance: The Health distribution has higher variance because 50% of the score comes from a **single variable** (Premature Death).

Skew: The heavy left tail represents counties with extremely poor health.

2. Community Conditions (Blue):

This Z-score is comprised of **24 variables** with varying weights. Some variables have higher weight, like Income Inequality and HS Completion which are weighted at 8% each, and others, like Dentists, are weighted at 1%.

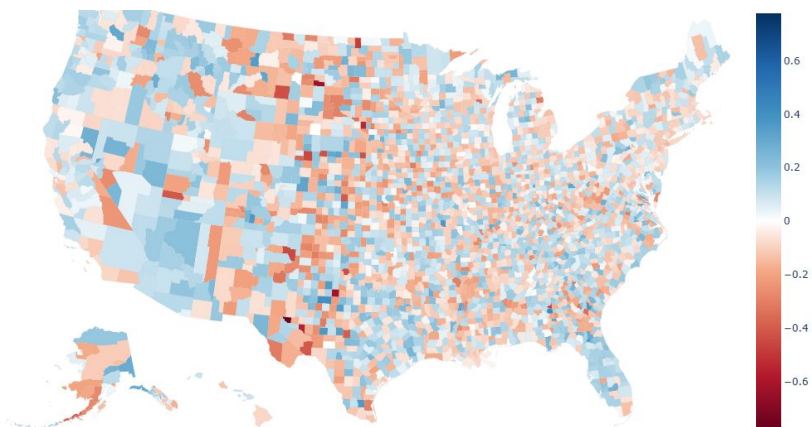
Central Tendency: The Community Conditions distribution is more compressed. Since it has 24 variables the extreme values tend to cancel each other out and result in a more normal distribution.



Data Manipulation Methods and Analysis

Mapping ACS Migration Data and County Health Rankings

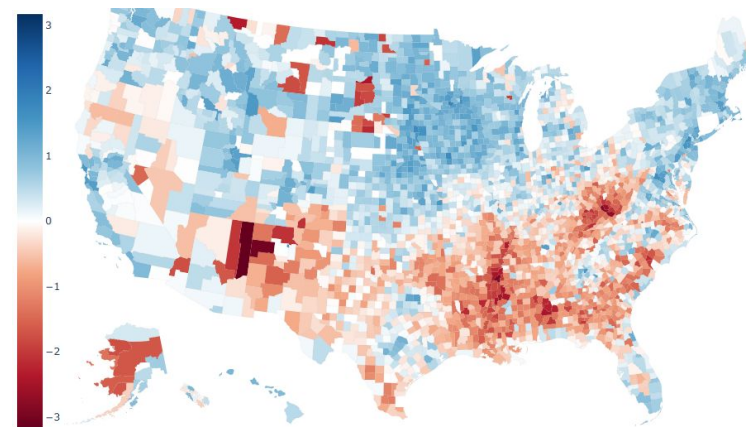
Intra-US Power-Transformed Net Migration Rate by County



Method: We applied a signed square root transformation to each county's net migration per capita in order to enhance the effectiveness of this visualization.

Analysis: Some geographic patterns appear, such as net out-migration from sparsely populated Great Plains counties and parts of the post-industrial Midwest, consistent with long-running patterns of limited local opportunity. However, when viewed alongside county health scores, migration patterns show little geographic structure.

US County Health Z-Scores



Analysis: In contrast to intra-US migration, county health scores show clear geographic structure. There are broad regions of good health in the upper Midwest, Rockies, and Northeastern and California coasts. Poor health in the Mississippi Delta is associated with long-standing economic disadvantage and limited healthcare access. Additional poor health clusters appear in Central Appalachia and in or around Native American reservations, where geographic isolation and persistent resource constraints contribute to worse population health*. These patterns suggest that community health conditions are more strongly shaped by place-based structural factors.

Insight: The contrast in geographic structure between migration and health patterns suggests that on a national scale, county health conditions alone are not a dominant driver of internal U.S. migration; rather, relocation decisions reflect an extensive and diverse set of economic, social, and personal considerations, with no region demonstrating consistent overall advantage.

* Source: [Validation of a New Continuous Geographic Isolation Scale](#) (Doogan et al., 2018, *Soc Sci Med*)

Analysis

Mapping Correlation Between Intra-US Migration and County Health

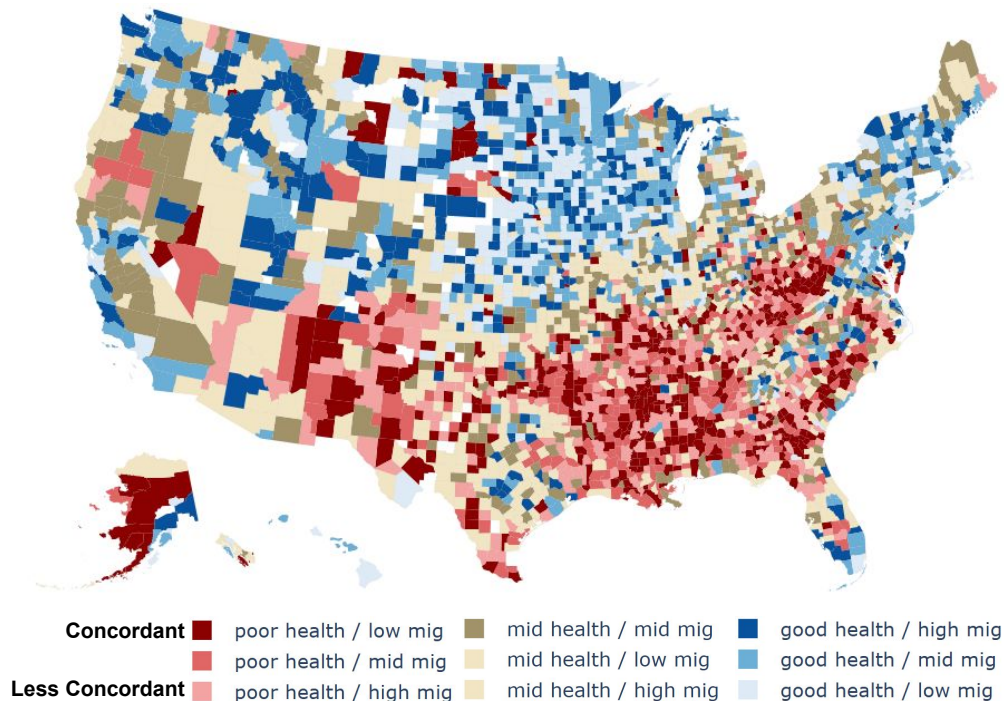
Method: This map is intended to illustrate the level of correlation between net migration per capita and county health z-scores. We split each county into one of three terciles for both variables. Counties with high net migration and good health, as well as counties with low net migration and poor health, are both shaded darker. This is intended to emphasize counties where net migration and population health are categorically correlated.

Insights

Overall Structure: Relative to intra-US migration, this map shows more geographic structure. Relative to county health scores, this map shows less geographic structure. This suggests that while the health-migration relationship is influenced by place-based structural factors, that relationship is far less deterministic than community health on its own.

Poor Health v. Good Health Counties: Looking at poor health and good health counties separately, we can see clear regions where poor health is categorically correlated to negative net migration. In contrast, counties where good health is categorically correlated to positive net migration are both more sparsely populated and less clustered. This suggests that the relationship between health and migration is notably stronger among poor health counties.

Poor Health Centers Remain: Although less regionally consistent, the aforementioned poor-health centers of the Mississippi Delta, Appalachia, and Native American reservations remain noticeable. Additionally, we can more clearly see poor health correlated with out migration in the rural Southeast.



Analysis

How do the aggregated health and community scores correlate to intra-US migration?

1. Linear Association (Pearson)

- **Health:** We observed statistically significant positive association ($p < 0.05$) with migration at the county level, but the effect size is extremely small (Pearson $r = 0.045$). This tells us that while a linear relationship exists it is very weak.
- **Variance Explained:** $R^2 = 0.002$, meaning **health Z-scores explain only 0.2% of the variance in migration.**
- **Community Conditions:** Our analysis found no statistically significant evidence of an association with net migration at the county level ($p = 0.468$), and these results were also observed when aggregating the data at the state level.

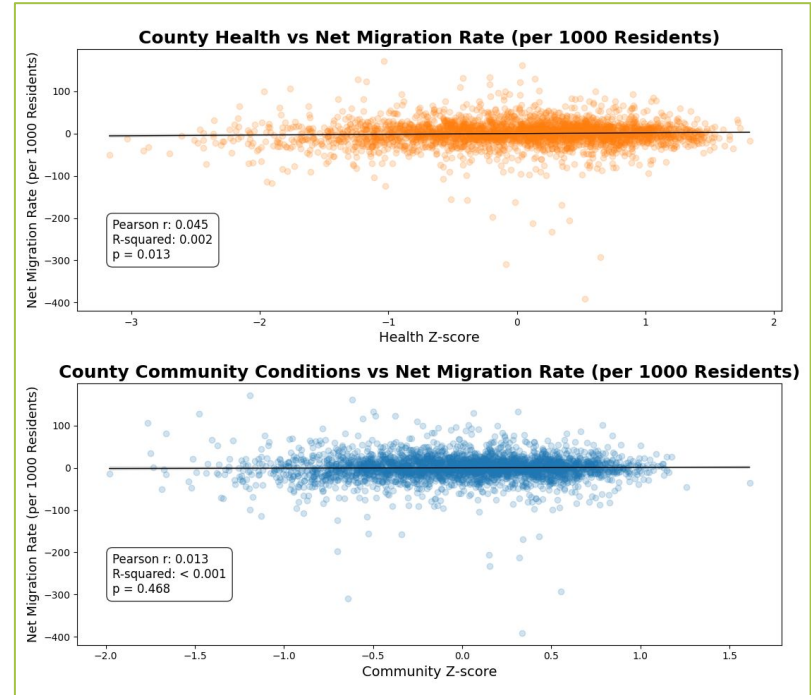
2. Monotonicity Test (Spearman)

We used Spearman's rank correlation coefficient, r_s , to test for monotonicity:

- **Results:** All r_s values were close to 0 for Health and Community Condition Z-scores at both the county and state levels with $p > 0.05$
- **Conclusion:** There is no statistical evidence of a monotonic relationship (i.e. migration does not consistently increase/decrease with rank).

Next Step

In the next slide we try a categorical approach to see if there is a relationship between Net Migration and Health/Community Condition Z-scores.



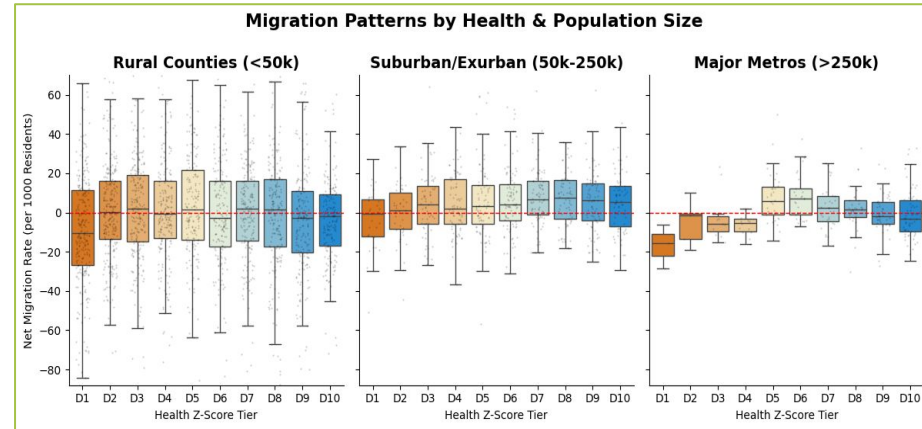
Analysis

Is there a relationship between net migration and county health and population?

Method: We split counties into 10 equal groups (Deciles) based on their Z-scores (D1 = Counties with lowest health, D10 = Counties with highest health) to categorize the Health and Community Conditions data.

Statistical Significance: A one-way ANOVA test (not shown) suggested significant differences in migration rates across health tiers, so we reject the null hypothesis that the mean migration rate is identical by score decile ($p < 0.001$, $F = 3.28$). This indicates that counties with different scores experience different average levels of net migration.

Box-and-Whisker Plot: To control for population we used `sns.catplot` to divide migration distributions by health decile and `sns.stripplot` to show the difference in sample size between the population categories.



Insights by Health Decile

Unhealthiest Counties: Strong Negative Net-Migration

Across all categories, counties with the lowest 10% health (D1) lose the most residents (7.6 people per 1000 residents on average). The net loss in the unhealthiest counties is **~20x higher** than that of the healthiest counties.

Average Health Counties: Positive Net-Migration

Counties with average health gain more migrants than any other group (between 1.8 and 3.9 people per 1000 residents on average).

High Health Counties: Low Negative Net-Migration

Counties with the highest 10% health lose an average of 0.37 people per 1000 residents on average.

Insights by Population

Rural (<50k): Weak Negative Association.

Beyond D1, mean net migration drifts from small gains (D2–D5) to losses (D6–D10).

Suburban (50k-250k): Positive Association.

As health scores improve net migration rates increase consistently.

Major Metros (>250k): Non-Linear Trend.

- **Low Health:** High out-migration.
- **Avg Health:** High in-migration.
- **High Health:** Diminishing returns/outflow could be driven by high cost of living.

Regression Analysis Steps and Initial Regression

How we used regression to identify predictors of migration

We wanted to use regression to identify which County Health Metrics correlate most strongly with migration rates (Net, In, and Out). We started by reducing the number of variables down from the 83 that resulted from merging migration data with the metrics.

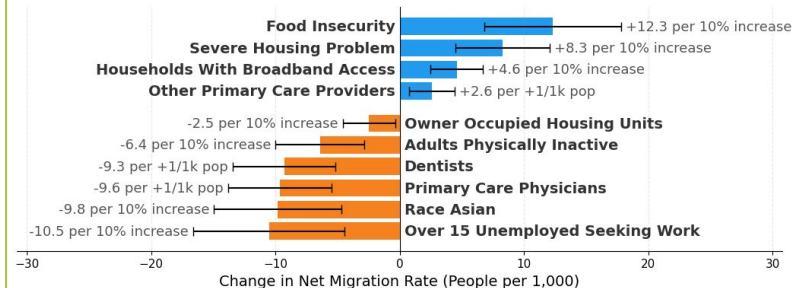
The first step was to manually remove duplicates (e.g., POP1YR and population_count) and highly correlated pairs (e.g., POP1YR and POP1YRAGO). We also needed to remove at least one variable from groups that added up to one - such as racial proportions. We also removed derived features like the Z-Scores and food_environment_index, which were derived from other features in the dataset.

Then, we wrote a Variance Inflation Factor function to identify variables which were highly multicollinear, and we removed those iteratively, rerunning VIF each time, until all VIF scores were under 10.

We then performed the following steps for each of the migration rates (Net, In, Out):

1. Run Scikit-Learn's LassoCV regression with default settings on standardized data to select the top-N most informative subset of the remaining features
2. Feed those features into Statsmodels' OLS model on the unscaled data
3. Review results, ignoring features that weren't statistically significant and omitting anything with an effect of less than 1 per 1,000 on the target migration rate
4. Visualize results in a bar chart using Matplotlib's barh function
5. Analyze results as top structural predictors of migration

Net Migration Top 10 Predictors (Scaled Impact per 1,000)



Our initial regression was against the Net Migration Rate. LassoCV selected 34 features, and OLS resulted in a R^2 value of 0.136, which is not out of the norm for cross-sectional survey data¹.

A low R^2 is also statistically expected. Net migration is an artifact of two distinct rates - In and Out - so noise compounds. A small change in migration - given 100 In, having 99 or 101 Out - can flip the sign (Net of +1 vs -1), which linear regression struggles to model.

With that in mind, we decided to look at the In and Out rates individually.

¹ Cohen (1988), *Statistical Power Analysis for the Behavioral Sciences*

Regression Analysis for Inward and Outward Migration

Running separate regressions led to increased clarity on key predictors of migration

The regression results and associated charts lead to distinct hypotheses about the structural factors associated with inward and outward migration.

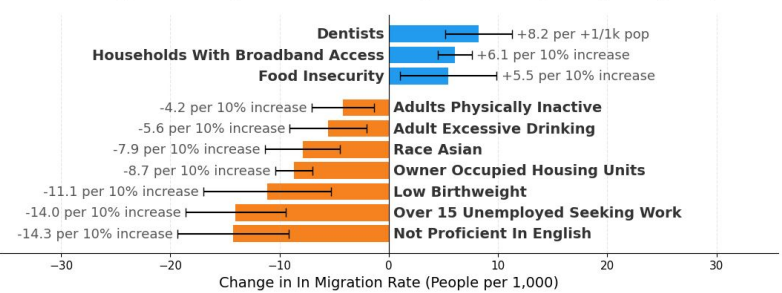
In Migration correlates strongly with proxies for infrastructure, opportunity, and housing availability.

- Broadband access serves as a strong predictor of in-migration, acting as a proxy for modernized infrastructure and commercial development.
- High unemployment, however, is strongly negatively correlated with inward migration.
- High rates of owner-occupied housing units are inversely associated with inward migration, as housing stock moves slower than rentals and has a higher barrier to entry (e.g., 20% down payment versus one month's rent), which limits availability.
- The positive correlation with food insecurity may indicate migration into more affordable areas, or it may capture transient, low-income populations like students and gig workers, who move frequently for other reasons.

Out Migration tells a different story. Some factors are in alignment - such as affordability - but there are also indicators of populations getting stuck.

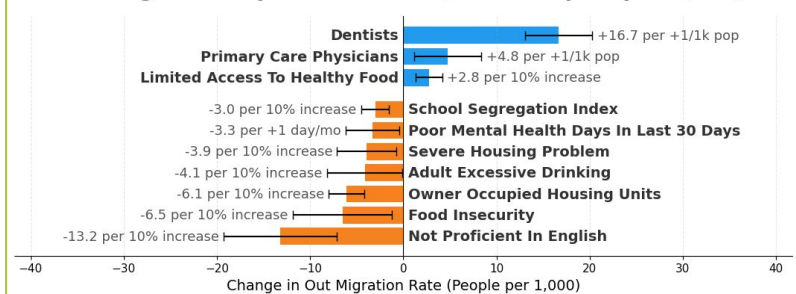
- Markers of affluent areas - such as Dentist and Primary Care Physician density - are among the strongest predictors of leaving.
 - Dentist density (a proxy for urban, service-rich areas) is highly correlated with both in- and out-migration. This indicates 'population churn', where urban centers have high levels of transient residents.
- Markers of poverty, however, show people getting stuck. As conditions (mental health, alcohol abuse, segregation, food insecurity) worsen, out-migration slows. This suggests a potential 'poverty-trap', where structural disadvantages deplete the financial or psychological resources required to relocate, though unobserved variables account for the remaining variance.

In Migration Top 10 Predictors (Scaled Impact per 1,000)



LassoCV selected 38 features for In migration, and OLS had an R^2 of 0.448 in this dataset, a strong fit by Cohen's conventions*.

Out Migration Top 10 Predictors (Scaled Impact per 1,000)



LassoCV selected 33 features for Out migration, and OLS $R^2=0.244$, which is moderate per Cohen*.

Statement of Work

Freya Van de Motter

Freya collected, cleaned, manipulated and initially explored the US Census Migration Dataset. She additionally took the lead on setting up the code environment (Deepnote) and establishing standards for readability and reproducibility across computational notebooks.

She conducted exploratory data analysis on the County Health Rankings merged with the Migration Flows using Sankey diagrams and histograms.

Freya merged datasets and took the lead on regression modeling by imputing missing values, using VIF calculations to identify and remove collinear variables. She then conducted Lasso regression to select important features which she fed into an OLS regression model of County Health Metrics against migration rates, with an in-migration R^2 of 44.8%, out-migration R^2 of 24.4%, and net R^2 of 13.6%.

She populated a fair share of slides in the final report.

Shay Gooding

Shay took the lead in problem formulation by defining our motivation—to learn more about the why behind people's migration decisions.

She also collected, cleaned, manipulated, initially explored the County Health Rankings Dataset, including important insights about rankings data distributions, and established the `functions.py` module to standardize the data ingestion and cleanings functions across team members.

She conducted initial exploratory analysis with a data distribution comparison using `pandas.melt` and `seaborn.displot`, then led the statistical analysis of the overall relationship by evaluating linear and monotonic associations using Pearson/Spearman correlations and R^2 analysis, a decile based ANOVA analysis, and a population controlled box-and-whisker analysis (`sns.catplot`).

She populated a fair share of slides in the final report.

Ransom MacLeith

Ransom identified the County Health Rankings dataset and conducted exploratory data analysis to confirm its potential as a counterpart to US migration.

He conducted initial research into existing studies on the influence of community health on migration decisions. He also collected, cleaned, manipulated and initially explored the County Health Metrics dataset. He created a config file for the Metrics dataset which was used to replace technical variable names (e.g., 'v139_rawvalue') with human readable names ('food insecurity').

Ransom merged datasets and conducted categorical correlation analysis using univariate and bivariate choropleth maps.

He populated a fair share of slides in the final report.

Project Collaboration: Collaboration went well, with strengths in repeatability and computational reproducibility. Team members conducted EDA within cohesive notebooks and created a separate Deepnote project to split development and master branches. We met at least once weekly, scheduling well with perfect attendance.

Improving Collaboration: The team will attempt to use Github in future projects to more safely and conveniently ensure proper data versioning and the storage of intermediate results.